PLANT PROTECTION

Plant Protection : These are measures put in place to protect our plants from harmful, environmental stressors and so on.

It encompasses the practices and measures used to safeguard plants from harmful organisms, environmental stressors, and other damaging factors. It involves:

OBJECTIVES

1. Preventing yield losses
2. Reducing crop damage
3. Protecting plant health
4. Minimizing environmental impacts
5. Ensuring food safety and security

SCOPE

1. Pest management (insects, mites, nematodes)
2. Disease management (fungal, bacterial, viral)
3. Weed management
4. Nematode management
5. Environmental stress management (drought, heat, cold)

METHODS

1. Cultural controls (pruning, sanitation, irrigation)
2. Biological controls (predators, parasites, beneficial micro-organisms.
3. Chemical controls (pesticides, herbicides, fungicides)
4. Resistant cultivars
5. Integrated Pest Management (IPM) strategies

IMPORTANCE

1. Ensures global food security.
2. Protects biodiversity
3. Maintains ecosystem health
4. Supports sustainable agriculture

5. Enhances human health and well-being.

Plant protection is a critical aspect of agriculture, horticulture, and forestry, as it helps maintain plant health, prevent yield losses, and ensure food safety and security.

Plant protection being the measures and strategies used to safeguard crops from pests, diseases, and environmental threats to ensure healthy growth and high yields, also includes ;

1. Pest Control : Managing insects, rodents, and other harmful organisms using biological, chemical or cultural methods.
2. Weed Management : Controlling unwanted plants that compete with crops for nutrients, water and sunlight.
3. Disease Management : Preventing and treating plant diseases caused by fungi, bacteria or viruses.
4. Integrated Pest Management (IPM) : A sustainable approach that combines biological, mechanical, cultural and chemical control methods.
5. Quarantine Measures : Restricting the movement of infected plants to prevent disease spread.
6. Use of Resistant Varieties : Growing crop varieties that are naturally resistant to pests and diseases.

Effective plant protection measures ensures food security, improves yields, and reduces economic losses in agriculture.

BRANCHES

1. Pathology
2. Entomology
3. Fires
4. Integrated Pest and Disease Management

PATHOLOGY

* Pathology is the scientific study of plant diseases, their causes, processes and effects on living organisms. How they occur (spread), how they can be controlled as well as avoided.
* In agriculture plant pathology focuses on understanding plant diseases caused by pathogens like fungi, bacteria, viruses and nematodes, as well as environmental factors such as drought or pollution.
* Plant pathology plays a crucial role in protecting crops, ensuring food security, and supporting sustainable agriculture.

BASIC CONCEPTS

* **Pathogen** : Any micro-organism or agent that causes disease in a living organism.
* **Phytopathogen (Plant pathogen)** : An organism that causes diseases in plants.
* **Biotic Diseases** : Diseases caused by <u>living</u> organisms such as bacteria, fungi, viruses and nematodes.
* **Abiotic Diseases** : Diseases or plant disorders caused by non-living (environmental) organisms or agents (factors) such as weather conditions (climate), soils, temperature, humidity, chemicals, etc.
* **Parasitism** : Is a type of biological relationship where one organism, called the <u>parasite</u>, lives on or inside another organism (host) and derives nutrients at the host's expense, often causing harm.
* **Epiphytes** : Are plants that grow on the surface of other plants (usually trees) merely for physical support but do not take nutrients from their hosts.

Unlike parasites, epiphytes rely on the surrounding environment for water and nutrients, often absorbing moisture from rain, air or organic debris.

* **Saprophytes**: These are organisms especially fungus (mushroom) that lives on decaying or decomposing dead organic matter such as decaying plants, animals, and other biological materials and depends on them for nourishment (obtains nutrients from them).
* **Pathogenicity**: Ability of an organism to cause a disease. It is not every attack by a pathogen will cause a disease. For example, not every mosquito bite will cause malaria. Therefore, pathogenecity is how powerful a pathogen is to cause a disease in a host.
* **Virulence**: The disease producing power of an organism. It is the degree of severity of the disease caused by a pathogen. It reflects how harmful or aggressive a pathogen is in causing damage to the host. It is often measured by factors such as
  * toxin production,
  * ability to evade the immune system, and
  * the extend of tissue damage caused.

  Pathogenicity = Whether an organism can cause disease.
  Virulence = How severe the disease it causes is.
* **Infectious Disease**: Diseases or disorders as a result of an infection by a plant pathogen. They are caused by pathogenic micro-organisms (eg bacteria, viruses fungi or parasites) They spread directly or indirectly from host to host.
* **Non-infectious Diseases**: Diseases as a result of abiotic diseases. They are caused by genetic factors, environmental influences, lifestyle choices or aging. The do not spread from host to host.
* **Aquired Resistance**: Resistance which is achieved in a

plant by innoculation or chemical application (spray). It is the ability of microorganisms to develop resistance to drugs or treatments that were previously effective.

* **Biotroph** : Plant pathogenic fungus that requires living hosts or cells for their survival. It requires a living host to grow and reproduce. It does not kill the host immediately but instead derives nutrients from living cells.
* **Hemibiotroph** : Pathogenic fungus that initially requires living cells for their survival but later kills the host and continues to live on the dead or decaying matters. It starts as a biotroph and later switches to necrotrophic
* **Infection** : Occurs when a pathogen enters a host, establishes itself, and begins to multiply, leading to disease development.
* **Inoculum** : It is the part of a pathogen (such as spores, bacteria or viruses) that initiates an infection when it comes into contact with a host.
* **Invasion** : Penetration and spreading of a pathogen in the host plant. Pathogen spreads within the host tissue after infection. It occurs through wounds, natural openings (stomata), or direct penetration.
* **Symptoms** : Are the visible or measurable effects of disease in a plant or animal caused by a pathogen. They are external as well as internal reaction or alterations of a plant as a result of a disease.
* **Incubation Period** : This is the time between the infection of a host and the appearance of symptoms. It is the period between transmission of an infection to its manifestation. It is measured in days. (eg A month is 30 days).
* **Sign** : Observable characteristics just at first sight

* **Disease Cycle** : It is the sequence of events in the life of a pathogen that leads to disease development and spread. It typically includes :
  1. Inoculation - the pathogen arrives on the host.
  2. Penetration - it enters the host
  3. Infection - it establishes and begins to grow
  4. Colonization / Invasion - it spreads within the host.
  5. Symptom expression - disease symptoms appear.
  6. Reproduction - The pathogen produces new inoculum (e.g spores, bacteria).
  7. Dissemination - the pathogen spreads to new hosts
  8. Survival - the pathogen persists until the next growing season.
* **Disease Syndrome** : Are a set of valid characteristics of a disease. It is a group of symptoms that consistently occur together in a disease, helping to identify the specific infection or disorder affecting a plant or organism.
* **Alternate Host** : This is different plant species that a pathogen requires to complete its life cycle. This means that even if one host is not available or enough it can still survive on others.
* **Collateral Host** : This is a plant species closely related to the primary host that can also harbor the pathogen but is not necessary for its life cycle. It helps in the survival and spread of the disease. It is a plant species that is a host to a number of pathogens.
* **Predisposition** : Refers to the factors that make a plant or organism more susceptible to disease. These factors can be genetic, environmental, or physiological. They make an organism to be prone to attacks.
* **Mutualism** : It is a symbiotic (beneficial association) relationship where two organisms benefit from each other.

Symbiosis is a close and long-term biological interaction between two different organisms, where at least one benefits. It can be beneficial, harmful or neutral for the organisms involved.

Types of Symbiosis

1. Mutualism : Both organisms benefit E.g Bees and flowers, (bees get nectar, flowers get pollinated).
2. Commensalism : One organism benefits, while the other is neither helped nor harmed.
3. Parasitism : One organism (Parasite) benefits, while the other (host) is harmed, e.g. ticks on animals (ticks feed on blood, weakening the host).

* Mutation ; Abrupt / sudden appearance of new characteristics in an individual as a result of accidental change in genes present in that individual. A permanent change in the DNA of an organism, which can lead to variations in traits, including resistance or susceptibility to disease. For example mutation in bacteria can lead to antibiotic resistance.
* Deficiency (In Plants) : Abnormalities that are observed as a result of lack or abnormal level of essential nutrients (too little or too much nutrients) In plant, deficiency occurs when essential nutrients are lacking, leading to poor growth and disease-like symptoms.

For Example :

1. Nitrogen deficiency - Yellowing of older leaves (Chlorosis).
2. Potassium deficiency - Brown scorching on leaf edges.
3. Iron deficiency - Interveinal chlorosis in young leaves

CONCEPT OF DISEASES IN PLANTS

* It is not yet known whether plants feel pain or discomfort like human beings because they do not talk or communicate.
* Therefore, it is difficult to pin point if a plant is diseased especially if there are no visible signs.
* Plant is considered to healthy or normal if it is able to carry out the following physiological functions:
  i. The meristematic cells are able to differentiate and divide.
  ii. They are able to to produce seeds, flowers or any other reproductive organs for survival and multiplication.
  iii. To carry out photosynthesis.

N.B: If a plant is unable to perform either one of the above functions or all of them even in the presence of favourable weather conditions, it is diseased

PLANT DISEASE

* This refers to series of visible or invisible response of plant cells due to pathogenic attack or environmental factors that will result in adverse changes in functions or integrity of a plant that leads to partial impairment or death of some of the parts of the plant or even the entire plant.
* The first step of plant disease identification and treatment is to have a definite and clear symptoms on a plant.

CLASSIFICATION OF PLANT DISEASE

Plant diseases are classified according to several parameters and some of them are as follows:

i. **Infected plant type**: What plant species are affected and which family.

ii Infected organ : Some diseases are perculiar only to one or two plants speaces.

iii. Disease symptom : Observe symptoms on plants may help to classify the plant disease.

iv. Type of phytopathogen : Certain phytopathogen does not exist in certain environments.

N.B : Identifying the phytopathogen is more useful criterion used for plant disease classification because it determines the disease cause, potential disease obligation as well as the measures of control to take.

CAUSES OF PLANT DISEASES

There are two ways in which pathogenic micro-organisms cause disease in plants:

1. They disturb metabolism of plants cells or tissues through enzymes and toxins and other substances that they secrete and by absorbing food stuff from the host cells for their own use.
2. Some pathogens may also cause diseases by growing and multiplying in the xylem and phloem tissue of a plant thereby blocking or disturbing pathway for water and mineral salts from the ground. Uptake of water and mineral salts from the ground through roots will be altered in a way that the plant may start wilting.

TYPES OF PLANT DISEASES

There are two types of plant diseases and these are

1. Infectious diseases
2. Non-infectious diseases

1. INFECTIOUS PLANT DISEASES

* Infectious plant diseases are also called Biotic plant diseases (or Biotic plant infections).
* They are caused either by fungi, bacteria, viruses, viroids or nematodes. These are living microorganisms, hence the name biotic infections.

2. NON-INFECTIOUS PLANT DISEASES

* Non-infectious diseases are also known as Abiotic diseases or infections.
* They are caused by too low or too high temperature, lack or excess soil moisture, lack or excess light, air pollution, lack of oxygen, nutrient deficiencies, mineral toxicities, soil acidity or alkylinity, etc.

LOSSES CAUSED BY PLANT DISEASES

i. Reduced quality and quantity of produce leading to reduced production hence shortage of food.
ii. Limit kinds of plants in an area.
iii. Some plant diseases make plant to be poisonous to humans as well as animals.
iv. Leads to economic loss for instance you spend a lot on your garden (investment) and its infested with a disease resulting in huge losses.
v. Leads to financial losses.
vi. Extinction of some plant species.

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DISEASE EPIDEMIOLOGY

* Diseas epidemiology is also known as Disease epiphytology. Hence, It is the study of the rate of disease or pathogen multiplication and the rate of spread in a plant population.
* Epidemiology deals with serious disease outbreaks.

IMPORTANCE OF EPIDEMIOLOGY

* It regulates the spread of diseases and put measures in place to mitigate the disease.
* In order to study how fast a disease spreads
* To facilitate disease forecasting or to predict what is likely to come.
* To formulate management strategies for instance to quarantine (control) the spread of diseases.

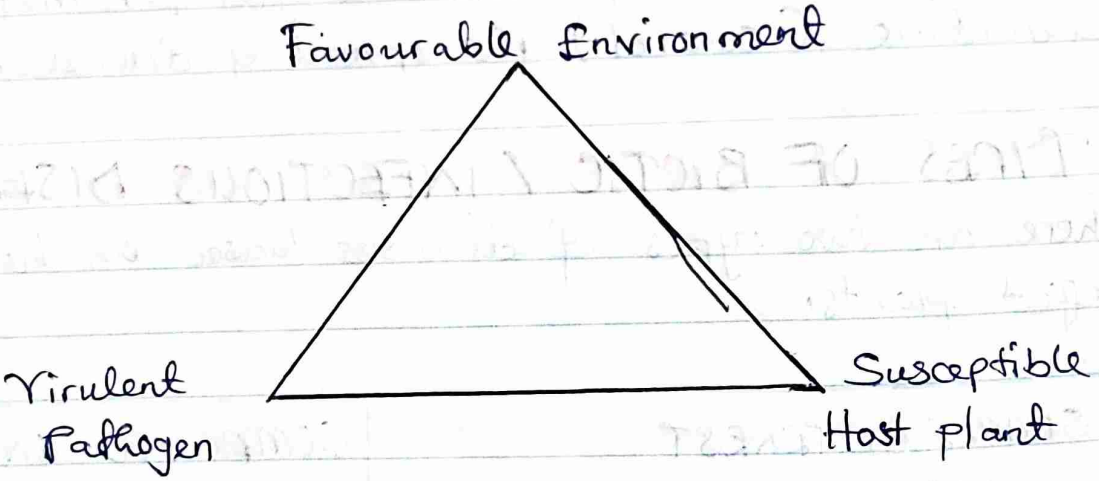
TYPES OF BIOTIC / INFECTIOUS DISEASES

There are two types of diseases based on how they affect plants.

SIMPLE INTEREST DISEASES (MONOCYCLIC)	COMPOUND INTEREST DISEASES (POLYCYCLIC)
Diseases have single cycle. It ends its cycle when it is controlled.	Diseases have many cycles. It re-occurs when you either control or not.
Spread slowly	Spread fast (rapidly)
Usually found in soil and water.	Usually spread by air eg fungal infections.
Have longer incubation period	Have shorter incubation period.

DISEASE TRIANGLE

* The Disease Triangle is a fundamental concept in plant pathology that explains the three essential factors required for a plant disease to occur.
* It is basically an interaction between three components of diseases that have been visualized in form of a triangle and generally referred to as a disease triangle.
* In order for a biotic disease to occur or an infectious disease to occur, there are certain conditions that need to be available.
* Each component or side of a triangle is represented by one of these components (factors).



In order for a disease to occur, the components must interact with each other.

1. PATHOGEN FACTOR

A disease-causing organism eg. fungi, bacteria, viruses.
- Virulence ; the ability to cause infection.
- Inoculum type ; the part of pathogen invading the host.
- Dispersal potential ; quick or slow spread of the infection.
- Host range
- Alteration between host (alterative host)

2. ENVIRONMENTAL FACTORS

Favourable conditions that allow the pathogen to impact and spread. These include:

i. Temperature ; each pathogen has an optimum temperature for growth and multiplication (development).

ii. Light ; plants that receive adequate light will have less pathogen invasion as the conditions around them will be dry and physiological processes in plant will be at their highest level e.g. photosynthesis. Too much watering of crops is ideal for the spread of diseases.

iii. Soil type ; Certain pathogens thrive well in loam soil, others clay soils and so on. Soils with low organic matter content are more preferred by pathogens.
Plants with nutrient deficiency are easily attacked.
Soils have solid particles (e.g. loam) which are closely packed, holding water for a long period of time without draining making it easier for penetration of pathogens.
Therefore, too much watering of crops in the garden is ideal for the spread of diseases.

iv. Soil pH ; Measure of acidity or alkylinity of the soil.
If the pH is lower and the acid is higher favours disease development. High pH = lower acidity also favours the development of a disease.
Soil pH is also an environmental factor because different soil pH favours different pathogens.

Moisture ; Cold weather, ideal environment for disease development.

Humidity ; Higher humidity, higher spread of diseases

Air pollutant ; It could either promote or discourage in influencing pathogen development by either aiding or

aiding or reduces the spread of diseases.

3. HOST PLANT FACTORS

A susceptible plant that can be infected.

i. Host Uniformity : Spread of diseases is mostly likely to affect **all** plants because of the same type of crop on the same field.

ii. Age of Host plant : Some pathogens prefer certain age of plants. Some are more common in younger than older and vice versa. For instance, if a certain disease which affects younger plants attacks old plants will not invade the host because it has already passed that stage.

iii. Season : Plants grow and produce in different seasons, so do pathogens thrive on plants in different seasons. For instance. Season development of pathogen attack perennial crops since they are there in all seasons (i.e rainy, dry and so on).

iv. Host resistance : Some plants ~~are~~ have resistance characteristics wile others are susceptible (vulnerable) to attack.

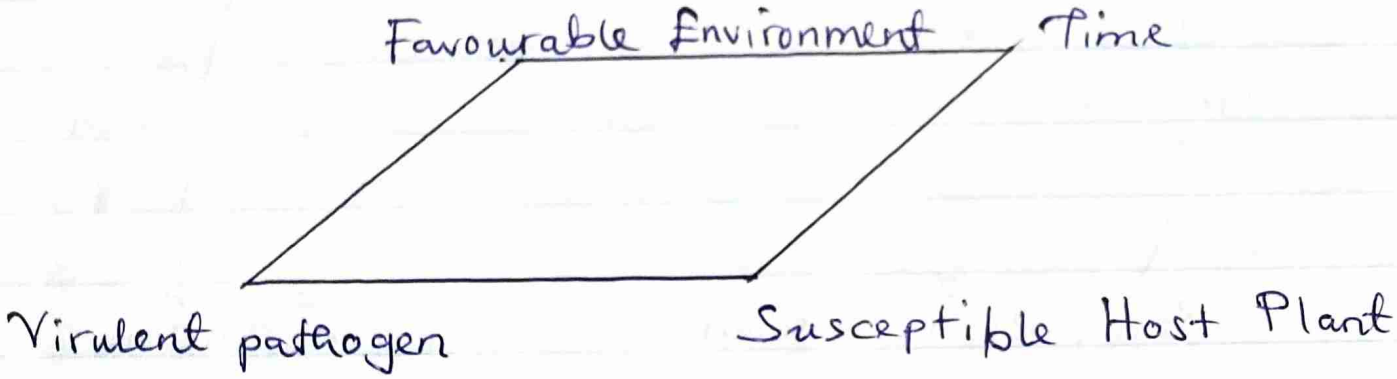
Plants differ according to genetic make up hence some are easily infected and others are not.

N.B: If any of these factors are missing, the disease will not develop. In absence of one or two factors of the disease triangle, will result in no disease development.

Managing plant diseases often involves disrupting one or more sides of the triangle, such as using resistant plant varieties, controlling pathogens or modifying environmental conditions.

4. TIME

* This is the fourth factor that has been observed to influence on the development of a disease.
* The amount of time that each of these three components described in the disease triangle and their interaction also have influence on disease development and this is called Disease Tetrahedron.
* Addition of time to disease triangle results in disease tetrahedron which is also known as Disease pyramid.



The effects of time on disease development becomes clear or apparent when we consider :

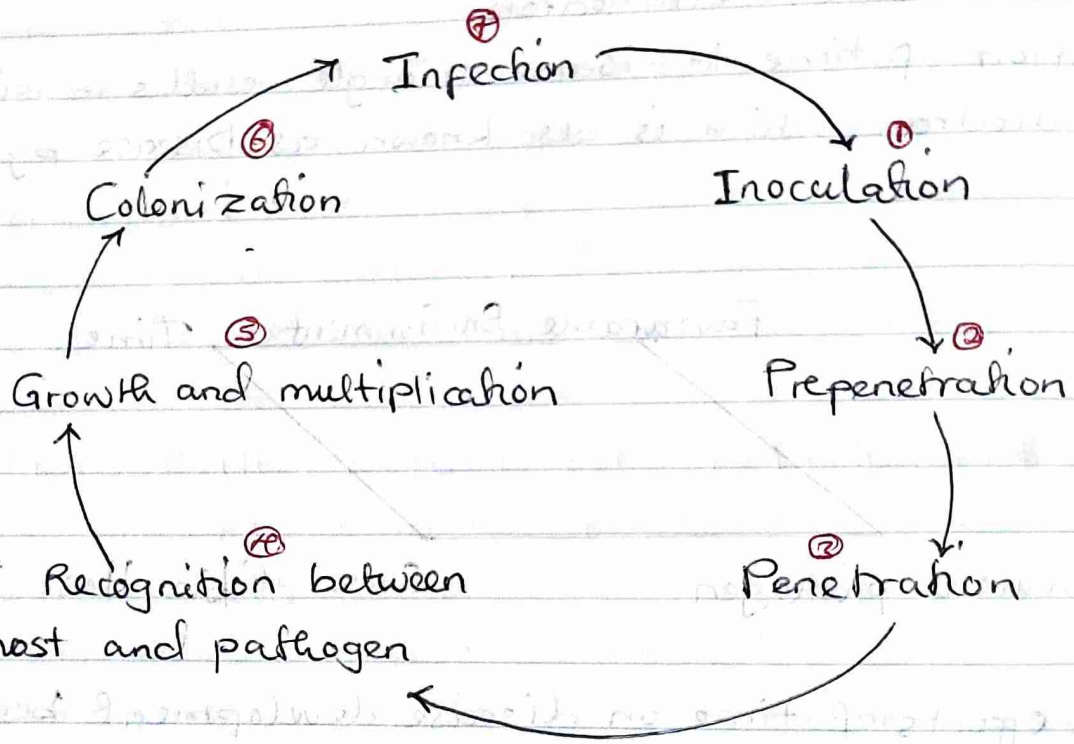
i. Time of the year (season)
ii. The duration and frequency of favourable temperatur
iii. The time of appearance of a vector.
iv. The duration of a cycle of a particular disease.

DISEASE CYCLE

A Disease cycle is a series of interconnected events leading to disease development.

It is the sequence of events in the life of a pathogen that leads to disease development and spread.

The disease cycle has the following stages :



1. **INOCULATION** : Transfer of pathogen inoculum to the infection site or part which is vulnerable to be infected. Entry points are infection points, and these infection sites can be wound (due to mechanical injury), natural opening (stomata) or hydathodes (opening of the bark of a tree), lenticils (opening on the roots) or adverse weather condition

Types Of Inoculum : There are two types of inoculum which are :

i. Primary Inoculum : This is the initial inoculum that lives dormant in other season (i.e winter or summer) and causes original infections in another season (Autumn/Spring

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ii. Secondary Inoculum : Is the one which is produced at the point of infection. These are actual infections which result from the primary inoculum.
Secondary penetrates to cause infections

2. PREPENETRATION

Prepenetration involves an attachment of pathogens to the host plant. The pathogens will either directly penetrate plants tissues through vectors (an organism carrying a pathogen which is a disease causing agent) or will attach through openings or the surfaces.
In other cases some pathogens will first make contact with the plant external surface and penetration process will take place, as they secrete certain substance which softens the host surface making it easier to penetrate or invade the host.

3. PENETRATION

It is similar to prepenetration. The pathogen will get to the plant system and establish in the cell or tissues while those attached to the surface will release some watery substances and dissolve the ourter part (surface) of the plant so that it becomes soft and easy to invade.
Penetration is easier when the environment is watery.

4. RECOGNITION BETWEEN HOST AND PATHOGEN

When a pathogen comes in contact with the host cell, the plant triggers a signal that either allows or rejects the pathogen growth and development.

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5. GROWTH AND MULTIPLICATION

Pathogen has been successfully implanted. At this the pathogen will start to multiply in the system of host.

6. COLONIZATION

Spread of the pathogen from one area to the other Pathogen has taken over the entire plant.

7. INFECTION

This is when a disease causing pathogen invades a plant and the symptoms are visible.

DISSEMINATION OF PATHOGEN

Dissemination (to spread or extend by dispersion) invo pathogen scattering, for growth and propagation, and spread abroad the entire host. Dissemination can be done by vari agents such as fungi, animals, water, human beings as well as infected material usage.

DEFENCE MECHANISMS

- Defensive mechanisms in plants against pathogens.
- It is very important to understand that whatever kind of defence or resistance a host plant employs against a pathog or against an abiotic agent is directly or indirectly controlled by genetic materials (genes).
- The following are some of the defence mechanisms that plant uses to defend themselves;

1. Non-host resistance: This is where a plant finds it easy to defend itself against a pathogen when its brought in contac with a pathogenic agent to which it is not a host.
2. Pre-existing defence of a plant against

2. Pre-existing defence structures : This is the first line of defence of a plant against pathogen on its surface (outward appearance) which the pathogen must adhere to and penetrate if its to cause an infection.

i. First pre-existing structure is the size, location and shapes of the stomata.
Some plants have more stomata located on the underside of the leaves hence making it difficult for pathogen to penetrate.
Some plants species have the stomatas which are very small hence making it difficult for penetration.

ii. The quality of wax ; plants releases wax (watery and sticky substances that plant secretes). The quality of wax form a water repellant surface to make it difficult for pathogen penetration.

iii. Thick cuticle ; increases resistance to infection in which the pathogen enters its host through direct penetration.

iv. Hairy leaf and stem surface ; the more hairy the leaves and the stem surfaces are, the better it becomes for plants to defend themselves against pathogens.

v. Presence of thorns ; thorns also deter pathogen vectors to cling or get in contact with the surface. Species with thorns can not be easily attacked.

3. Pre-existing chemical defences ; A particular pathogen will not infect certain plant varieties even though no structural barriers are present (exists). This clearly suggests that defence mechanisms of a chemical are responsible for that kind of resistance and these are ;

i. Inhibitors released by plant to its environment : Certain plant species release unpleasant chemical compounds to its environment thereby deterring (prevent) the vectors

ii Inhibitors present in plant cells before infection : Some plants are resistant to diseases caused by certain pathogens because of one or more inhibitory antimicrobial compounds known as phytoanticipins which are present in the cell before infection. This is also genetic

4. Defence through lack of essential factors

i. Lack of recognition between the host and pathogen : If the pathogen does not recognize the plants as one of its host may not become attached to the plant .... Process of disease cycle won't start and there will be no infection.

ii. Lack of essential substances for the pathogen : Plants that for some reason do not produce one of the substances essential for the survival of the pathogen would be resistant to the pathogen that requires that particular substance for survival.

5. Induced Structural Defence Mechanism

Plants usually respond by forming one or more types of structures that are useful in defending themselves from other pathogen invasion even after penetration through the wounds and natural openings like stomata and so on.

i. Cytoplasmic difference reactions

This is a defence structure that is formed when the cytoplasm of the cell under attack becomes dense and disintegrates the pathogen.

ii. Cell Wall Defence structure

Just like the cytoplasm, this involves the cell wall of the involved (invaded) cells which undergoes morphological changes and disintegrates the pathogen.

iii. Hypersensitive Defence Reaction

This is also known as necrosis (necrotic) defence reaction. Hypersensitive (more than active)

This is where the invaded cell of the plant will

protect the other parts of the plant from further invasion by dying.

iv. Formation of Abscission layers

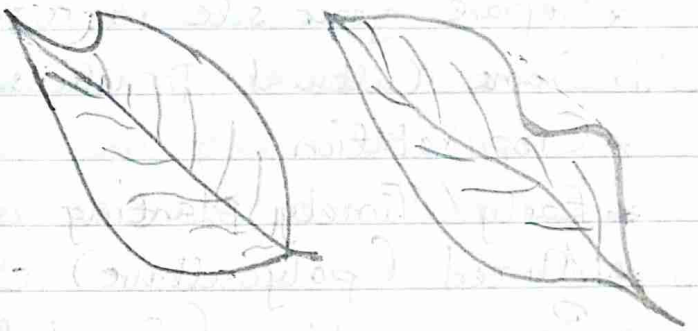
Abscission : Act, process or state of being cut off.

* Shedding of flowers and leaves and fruit following formation of scar tissue in a plant.
* Formation of abscission layers simply means there is a layer that is formed on the young and active leaves of some plants after infection. The plant will discard (no longer of service) the infected area along with a few uninfected cells thereby protecting the rest of the leaf tissue from being invaded.

Before infection



After infection



v. Deposition of gums

* After the cell is infected, plants will trigger the formation of gums and they are deposited between the intercellular spaces (spaces existing between two cells) thereby making it difficult for the pathogen to move from one cell to another. The pathogen therefore becomes isolated, starves and sooner or later dies.

6. Genetically Engineering Disease Resistant Plants

* Scientists have isolated a number of plant genes for resistance (R-genes) and this rapidly becoming common across the world.
* Scientists have isolated and make certain plant species more resistant to certain pathogens.

7. Cultural Practices that aim at reducing plant diseases
Cultural practices are basically practices or routines undertakings that are done at the farm in order to discourage or to create unfavourable conditions for the pathogen not to attack. Some of these practices are proper weeding, timely planting, crop rotation, proper fertilization and so on.

CULTURAL PRACTICES AND CONTROL MEASURES

CULTURAL PRACTICES

i. Site selection and preparation
* Always grow your plants in well drained soils as they retain moisture or water for some time in order to facilitate crop production.
* Prepare your site in a manner that it is clear and clean

ii. Some Cultural practices to be conducted
* Crop rotation
* Early / Timely planting
* Mixed (polyculture) cropping
* Proper weeding (Sanitation on the field)
* Proper fertilizer application (Neither too little nor too much)
* Sanitation (Tools, store room has to be clean)
* The use of resistant and tolerant varieties
The resistance in the varieties used at the farm, for instance the seeds, will result in the crop not to be infected whereas tolerant characteristic will result in the crop not be harmed in spite of being infected compared to other produce (these as a result of any variety used).

iii Use of propagative

DISEASE CONTROL MEASURES

* Proper use of chemicals is encouraged as a control measure. Some chemicals used are:
  i. Insecticides
  ii. Herbicides
  iii. Fungicides
  iv. Pesticides
* Inspection and quarantine (measures put in place)
  Rigorous (severe or intense) inspective measures have been put in place by authority to ensure that only seeds and plant products that meet the standards are allowed to reach the market, entry into the country or region or taken outside the country.
* Crop certification
  - It involves compulsory inspection systems that have been put in place in various countries to inspect and issue certificates for seeds and plant products.
  - Only those that have been certified and tested will reach the market and stiff penalties are in place for those that evade (avoid) the testing and certification process. Some companies that certify seeds; Seed Certification Institute (SCI), ZARI

COMMON PLANT DISEASES AND THEIR SYMPTOMS

1. FUNGAL DISEASES

1.1. INTRODUCTION

* Fungi are eukaryotic organisms, meaning that their cells contain membrane bound organelles and clearly defined nucleus.
* They are spore bearing organisms and do not have chlorophyll (the green pigment). Hence they are not classified as plants but exist as a kingdom on its own.
* Fungi are generally small and microscopic and in most cases they appear in filamentous form, branched form as they are spore bearing organ that lie harmlessly.
* Fungi play essential roles in ecosystems as decomposers, symbionts (live close to their partner organism, depend on each other for survival and can be found in all kingdoms of life) and pathogens.
* Some fungi known as obligate parasite (Biotroph) can grow and multiply only by remaining in association with the host plant in their entire life. While others are called Non-obligate parasites, they can grow and multiply on dead / decaying organic matter or on the living host.

1.2. REPRODUCTION

* Fungi reproduce by means of spores. Spores are reproductive bodies that consists of one or a few cells that are formed asexually but sometimes produce sexually.
* They reproduce both sexually and asexually through spores, which can be dispersed by wind, water or animals. However, there are few fungi that can reproduce sexually.
* The asexual spores are produced inside a sac referred to as Sporangium.

* Some spores are able to move on their own (in short distance) by use of the fragella (singular: fragellum), which is a tail-like structure.
* The spores that have a fragellum are referred to as Zoospores.
* Sexually reproduction occurs in some groups of fungi where two cells of similar size which are referred to as Zygomycetes unite and produce a zygote and that particular zygote is referred to as Zygospore.

1.3. CHARACTERISTICS

Eukaryotic Organisms: Fungi have complex cells with a nucleus and membrane-bound organelles.
Heterotrophic: They cannot produce their own food through photosynthesis. Instead, they absorb nutrients from their environment.
Cell Walls: Unlike plants (which have cellulose cell walls), fungal cells are made of chitin, a tough, flexible polysaccharide.
Habitat: Fungi are found in almost every environment, including soil, water, air, and on or inside other organisms.

1.4. MORPHOLOGY

* This is the structure of fungi. The external (outward) appearance.
* Most fungi have a filamentous like body structure called mycelium branches out in all directions and the group of mycelium are called hyphae.
Hyphae: The basic structural units of fungi are thread-like filaments called hyphae. These grow and branch to form a network called a mycelium.

Mycelium: The mycelium is the vegetative part of the fungus, responsible for nutrient absorption and growth.
Fruiting Bodies: In some fungi, the mycelium produces reproductive structures like mushrooms, puffballs, or molds, which release spores.

1.5. DISSEMINATION

* Zoaspores are the only kind of fungi that can move by themselves. However, they can only move for shorter distances like few millimeters or centimeters only.
* The other types of fungi that are unable to possesse a fragella disseminate by means of insects, plants, wind, animal agents and more long distances. They are called vectors.

1.6. TYPES OF FUNGI

Fungi are classified into several groups based on their reproductive structures and life cycles:

Ascomycetes (Sac Fungi):
* Produces spores in sac-like structures called asci.
* Examples: Yeasts, morels, truffles, and the fungi causing powdery mildew and Dutch elm disease.

Basidiomycetes (Club Fungi):
* Produce spores on club-shaped structures called basidia
* Examples: Mushroom, puffballs, rusts and smuts.

Zygomycetes:
* Reproduce through zygospores
* Examples: Bread molds (Rhizopus).

Chytrids:
* Aquatic fungi with flagellated spores.
* Some are pathogens of amphibians, causing disease like chytridiomycosis.

Deuteromycetes (Imperfect Fungi):
* Fungi that do not have a known sexual reproductive stage.
* Examples: Penicillium (source of Penicillin) and Aspergillus (Human pathogens

1.7. ROLES OF FUNGI
Decomposers:
* Fungi break down dead organic matter, recycling nutrients back into the ecosystem.
* Examples: Saprophytic fungi like Mucor and Rhizopus.
Symbionts:
* Fungi from mutualistic relationships with other organisms.
    * Mycorrhizae; Fungi that associate with plant roots, enhancing nutrient uptake.
    * Lichens; Symbiotic associations between fungi and algae or cyanobacteria.
Pathogens:
* Some fungi cause diseases in plants, animals and humans.
    * Plant pathogens: *Phytophthora*, *Fusarium*, *Puccinia*.
Industrial and Medical Uses
* Fungi are used to produce antibiotics (e.g. penicillin from *Penicillium*), enzymes, and food products like cheese, bread, and beer.
* Yeasts (*Saccharomyces cerevisiae*) are used in fermentation and biotechnology.

N.B: Asexual fungi: Myxomycetes and Oomycetes
Sexual fungi: Zygomycetes.

1.8. IMPORTANCE OF FUNGI

Ecological Balance : Fungi are critical for nutrient cycling and soil health.
Agriculture : Some fungi improve plant growth (mycorrhiza) while others cause crop diseases.
Medicine : Fungi are sources of antibiotics, immunosuppressants and cholesterol-lowering drugs.
Food industry : Fungi are used in producing fermented foods, cheeses, and beverages.
Bioremediation : Fungi can break down pollutants and clean up contaminated environments.

1.9. SIGNS AND SYMPTOMS OF FUNGAL INFECTION

* Fungal diseases are among the most common and destructive plant diseases, affecting a wide range of crops and ornamental plants.
* In general, fungi causes local or general necrosis (whole part turning yellow and dying) and this will lead to reduced (stunted) growth of a plant.

N.B : Local (necrosis) - whole part of a plant is affected.
Delocalize - only part of a plant is affected.

* Some of the plants grow rapidly and this is called hypertrophy.
* Some of the signs and symptoms include the following :

i. Mildew : Whitish / Brownish powder-like substances that cover the leaves.

a. Powdery Mildew

- Symptoms : White or gray powdery spots on leaves, stems, and flowers.
  - Leaves may curl, yellow and drop prematurely.
- Cause : Caused by fun

(e.g. Podosphaera, Erysiphe). Thrive in warm, dry climates with high humidity.

- Treatment: Apply fungicides like sulfur or potassium bicarbonate.
  - Improve air circulation around plants.
  - Remove and destroy infected plant parts
  - Use resistant plant varieties.

b. Downy Mildew

- Symptoms: Yellow or pale green patches on upper leaf surfaces, with a fuzzy, grayish growth on the underside.
  - Leaves may wilt and die.
- Cause: Caused by Peronospora or Plasmopara species. Favors cool, wet condition
- Treatment: Apply copper-based fungicides.
  - Avoid overhead watering.
  - Remove infected plants and debris (Decay matter)
  - Rotate crops to prevent recurrence.

ii. Root Rot: Decaying of a part or whole root system of a plant.

- Symptoms: Wilting, yellowing leaves, and stunted growth. Roots appear brown, mushy and decayed.
- Cause: Caused by fungi like Phytophthora, Pythium, or Rhizoctonia
  - Overwatering and poor drainage contribute to the problem.
- Treatment: Improve soil drainage
  - Avoid overwatering
  - Use fungicides like thiophanate-methyl
  - Remove and destroy severely infected plants.

iii. Damping - Off : It is common in seedlings like carrots, cucum
- Fungi prefer moist places (damp places) with low temperatures.

- Symptoms : Seedlings collapse and die shortly after germination.
  - Stems appear water-soaked and thin.
- Cause : Caused by fungi like Pythium, Rhizoctonia or Fusarium.
  - Common in overly wet, poorly drained soil.
- Treatment : Use sterile potting mix
  - Avoid overwatering
  - Apply fungicides to soil before planting
  - Ensure proper air circulation.

iv. Blights : General and extremely rapid brown and deaf

a. Late Blight :
- Symptoms : Dark, water-soaked lesions on leaves and stems. White fungal growth may appear on the underside of leaves in humid conditions.
- Cause : Caused by Phytophthora infestans. Spreads rapidly in cool, wet weather.
- Treatment : Apply fungicides containing copper or mancozeb
  - Remove and destroy infected plants
  - Avoid overhead watering.
  - Rotate crops and use resistant varieties.

b. Botrytis Blight (Gray Mold)
- Symptoms : Grayish-brown fuzzy mold on flowers, leaves and stems. Affected tissues may rot and collapse.
- Cause : Caused by Botrytis cinerea. Thrives in cool, humid conditions.
- Treatment : Remove and destroy infected plant parts.
  - Improve air circulation and reduce humidity.
  - Avoid overcrowding plants and apply fungicides like chlorothalonil.

v. Wilt : Also comes after spots, blights, curls. It is loss of turgidity (swollen) and generalized group of leaves.

a. Verticillium Wilt

- Symptoms : Yellowing and wilting of leaves, often starting on one side of the plant. Brown streaks in the vascular tissue.
- Cause : Caused by Verticillium dahliae or Verticillium albo-atrum. Soil-borne fungus that infects roots
- Treatment : Use resistant varieties.
  - Solarize soil to kill pathogens.
  - Rotate crops with non-host plants.
  - Remove and destroy infected plants.

b. Fusarium Wilt

- Symptoms : Yellowing and wilting of leaves, often starting on one side of the plant. Brown discoloration in the vascular tissue.
- Cause : Caused by Fusarium oxysporum. Soil-borne fungus that infects roots and spreads through the plant.
- Treatment : Use disease-free soil and resistant varieties.
  - Solarize soil to kill pathogens.
  - Avoid overwatering and ensure proper drainage.
  - Remove and destroy infected plants.

vi. Rust

- Symptoms : Orange, yellow, or brown pustules on leaves and stems. Leaves may turn yellow and drop.
- Cause : Caused by fungi in the order Pucciniales (e.g. Puccinia, Uromyces).
  Spreads through wind and water.

- Treatment: Apply fungicides like neem oil or sulfur.
  - Remove and destroy infected plant material.
  - Ensure proper spacing for air circulation.
  - Use resistant varieties.

vii. Leaf spots: Localized lesions consisting of dead or collapsed cells.

viii. Die bark: Excess necrosis (whole) of the leaf tip.

ix. Galls: This is the enlarged portion of a plant organ (roots, fruits, leaves etc).

x. Witches Brooms: Mass of growth on a plant. Outward branching of twigs.

xi. Leaf curl: Leaf edges fold inward. Curling is accompanied by dropping off of leaves.

xii. Cankers:

PREVENTION AND MANAGEMENT

1. Cultural Practices
   * Rotate crops to reduce soil-borne pathogens.
   * Use disease-resistant plant varieties.
   * Avoid overhead watering to reduce leaf wetness.
   * Ensure proper spacing for air circulation.
2. Sanitation
   * Remove and destroy infected plant material
   * Clean tools and equipment to prevent spread.
3. Chemical Control
   * Apply fungicides as a preventive measure or at the first sign of disease.
   * Use organic options like neem oil, sulfur, or copper-based products.
4. Environmental Management
   * Maintain proper humidity and temperature levels
   * Improve soil drainage to prevent waterlogging.

DATE: / /

2.0. BACTERIAL DISEASES

2.1. INTRODUCTION

Bacteria are generally single-celled microorganisms whose genetic material (DNA) is not bound by a membrane and therefore is not organized into a nucleus.

Most bacteria are strictly saprophytic and as such are beneficial to humans because they help decompose the enormous quantities of organic matter produced yearly by humans, animals and factories as waste products or by the death of plants and animals. Several species cause diseases in humans such as tuberculosis, pneumonia, and typhoid fever, and a similar number cause diseases in animals, such as brucellosis and anthrax.

Bacteria are in different shapes, some are rod shaped, spherical, spiral, or filamentous (thread-like). Some bacteria can move through liquid media by means of flagella, whereas others have no flagella and cannot move themselves.

2.2. CHARACTERISTICS OF PLANT PATHOGENIC BACTERIA

2.2.1 Morphology

Majority of the plant pathogenic bacteria are rod-shaped.

The cell walls of bacteria of most species are enveloped by a viscous layer forming a definitive mass around the cell; is called a capsule.

All the material inside the cell wall constitutes the protoplast.

Rod-shaped phyto-pathogenic bacteria reproduce by the

asexual process known as binary fission, or fis
(splitting, or breaking up into parts).

This occurs by the inward growth of the cytoplasmic membrane toward the center of the cell, forming a transverse membranous partition dividing the cytoplasm into ~~the~~ two approximately equal parts.

The two layers of the cell wall material, are then synthesized between the two layers of the membran

When the formation of these cell walls is completed, the two layers separate, splitting the two cells apar

Bacteria reproduce at an astonishingly rapid rate. Under favorable conditions, bacteria may divide every 30 to 50 minutes, one bacterium becoming 2 becoming 4, 4 becoming 8, 8 becoming 16 and so on. Over a million bacteria can be produced in a day.

2. Transmission and Infection

In order for a bacterium to produce a disease in a plant, the bacterium must first invade the plant tissue and multiply. Bacterial pathogens enter plant through wounds, principally produced by adverse weather conditions, humans, tools and machinery, insects, and nematodes, or through natural openings such as stomata, lenticels (pores in the stem for gaseous exchange), hydathodes, nectar-producing glands, and leaf scars.

Most foliage invaders are spread from plant to plant by wind blown, rain or dust. Humans disseminate bacteria through cultivation, grafting, pruning and transporting diseased plant material. Animals, including insects and mites, are other common transmission agents.

When conditions are unfavourable for growth and multiplication, bacteria remain dormant on or inside plant tissue. Some, such as the crown gall bacterium, may survive for months or years in the soil.

Bacterial diseases are influenced greatly by temperature and moisture. Often, a difference of only a few degrees in temperature determines whether a bacterial disease will develop. In most cases, moisture as a water film on plant surfaces is essential for establishing an infection.

2.2.3. Ecology And Spread

Almost all plant pathogenic bacteria develop mostly in the host plant as parasites, on the plant surface, especially buds, as epiphytes, and partly in plant debris (remains) or in the soil as saprophytes.

The main classes of bacteria that causes diseases in plants are agrobacteria which are rod-shaped and possess flagella, clavibacter which are mainly non-motile (incapable of moving by themselves), erwinia which are straight rods and are motile by means of flagella and the last group is pseudomonas which are straight or curve shaped and are most abundant in soils, freshwater and marine environments. Others are ralstonia, xylella and xanthomonas groups.

2.4. Classes of Bacteria Causing Diseases

i. Agrobacteria (rod-shaped with flagella)
ii. Clavibacter (non-motile)
iii. Erwinia (straight-rod and motile)
iv. Pseudomonas (straight or curve shaped).
v. Xanthomonas.

2.3. SYMPTOMS AND SIGNS CAUSED BY BACTERIAL DISEASES

Bacterial diseases can be grouped into four broad categories based on the extent of damage to plant tissue and the symptoms that they cause, which may include:

i. Vascular wilt

Vascular wilt results from the bacterial invasion of the plant's vascular system. The subsequent multiplication and blockage prevents movement (translocation) of water and nutrients through the xylem of the host plant. Drooping (bending downward), wilting (lose freshness), or death of the aerial plant structure may occur; examples include bacterial wilt of sweet corn, tobacco, tomato, and cucurbits (e.g., squash, pumpkin and cucumber).

ii. Necrosis

Necrosis (= Localized death of living cells).
Pathogens can cause necrosis by secreting a toxin (poison). Symptom include formation of leaf spots, stem blights or cankers.

iii Soft rot

Soft rot diseases are caused by pathogens that secrete enzymes capable of decomposing cell wall structures, thereby destroying the texture of plant tissue i.e. the plant tissue becomes macerated (soft and watery). Soft rots commonly occur on fleshy vegetables such as potato, carrot, eggplant, squash, and tomato.

iv. Tumors

Tumor diseases are caused by bacteria that stimulate uncontrolled multiplication of plant cells, resulting in the formation of abnormally large structures.

DATE: / /

2.4. EXAMPLES OF SOME SPECIFIC BACTERIAL DISEASES

Below is a list of examples of some common bacterial diseases

i. Bacterial spots and blight

These appear as spots of various sizes on leaves, stems, blossoms, and fruits. In some bacterial diseases the spots continue to advance rapidly and the diseases are then called blights.

In severe infections the spots may be so numerous that they destroy most of the plant surface and the plant appears blighted, thus producing large areas of dead plant tissue and blighted plants.

In dicotyledonous plants the bacterial spots on some hosts are restricted by large veins, and the spots appear angular. For the same reason, bacterial spots on monocotyledonous plants appear as streaks or stripes.

a. Bacterial leaf spot on bean
b. Bacterial spot (xanthomonas) on citrus leaves.
c. Tomato bacterial blight
d. Bacterial blight in bean
e. Galls on fruit trees (Tumefaciens)
f. Crown gall on fruit tree (Agrobacterium

2.5. MANAGEMENT AND CONTROL OF PLANT BACTERIAL DISEASES

Bacterial diseases of plants are usually very difficult to control. Frequently, a combination of control measures is required to combat a given bacterial disease. The following measures can help to control bacterial as well as other plant diseases:

Chemical controls : Involve the use of chemicals like fungicides, bactericides to control plant disease. Chemical controls can be effective, but it is important to properly identify the problem which may require submitting samples to a diagnostic lab for testing.

Genetic controls : Involve breeding disease resistant plants which creates new cultivars or varieties that perform better.

Cultural practices : This involves the use of various practices that aim at avoiding the provision of conducive environment for the survival and multiplication of the pathogen. For example, crop rotation to disrupt year-to-year pest cycles, sanitizing tools after using them on diseased plants to prevent further spread, planting warm seasoned crops after the soil has warmed to prevent root rot, watering plants in the morning so they can dry during the day and remain dry through the night to prevent fungal, and bacterial diseases, and pruning and trimming plants in such a way that it promotes airflow and more light penetration to prevent infection

Regulatory controls : These are based on exclusion of pathogenic organisms, highly susceptible host plants, and alternate host plants from a particular area.

Biological controls : This essentially involve using nature to prevent or treat diseases.

Sanitation practices : Aiming at reducing the inoculum (part of pathogen invading host) in a field by removing and burning infected plants or branches, and at reducing the spread of bacteria from plant to plant. This includes cleaning of the farm equipment and proper weeding.

DATE: / /

Adjusting Fertilizing and Watering: This is done so that the plants are not extremely succulent (juicy) during the period of infection may also reduce the incidence of disease.

Crop Rotation: This can be very effective with bacteria that ~~reduce~~ have a limited host range, but is impractical and ineffective with bacteria that ~~have~~ can attack many types of crop plants.

Seed Treatment: With hot water at about 50°C (120°F) is also effective against management of bacterial diseases in carrots, eggplant, pepper and tomatoes.

Destruction: Of all plant materials that may harbour (shelter) pathogenic fungi.

Biosecurity measure are also important to avoid the transfer of disease pathogens from an infected area to another. Biosecurity measures, practices or measures put in place to prevent or mitigate the risk of introducing pathogenic agents into, or spread within, or release from crop production site (farm).

#. Quarantine (Isolation) or, as it is usually called, purification of viruses is a very effective method of managing viral pathogens.

3.0 VIRUSES

3.1 INTRODUCTION

A virus is a nucleoprotein that multiplies only in living cells and has the ability to cause disease.
It is too small to be seen individually with a light microscope.
All viruses parasitize cells and cause a multitude of diseases in all forms of living organisms.
Virus consist of nucleic acid and protein, with the protein forming a protective coat around the nucleic acid. Although viruses can take any of several forms they are mostly rod shaped.

3.2. CHARACTERISTICS OF PLANT VIRUSES

From the onset, it is important to note that plant viruses differ greatly from all other plant pathogens not only in size and shape, but also in the simplicity of their chemical constitution and physical structure, methods of infection, multiplication, translocation within the host, dissemination, and the symptoms they produce on the host.
Because of their small size and the fact that they are transparent, viruses generally cannot be viewed and detected by the methods used for other pathogens.
Viruses are not cells nor do they consist of cells. Meaning that they can or are both living organism (if they are in living cells) and non-living organism if they exhist on their own (as they cannot survive for a long period of time without a host). Non-living because they are a material (protein)

DATE: / /

3.3. DETECTION

Viruses are very difficult to detect and to see under light microscope.

The present methods of detecting plant viruses involve primarily the transmission of the virus from a diseased to a healthy plant by budding, or grafting, or by rubbing leaves of healthy plants with sap (cell sap) from an infected plant.

Certain other methods of transmission, such as insect vectors, are also used to demonstrate the presence of a virus.

Most of these methods, however, cannot distinguish whether the pathogen is a virus or bacterium. Therefore, the most definitive proof of the presence of a virus in a plant is provided by purification, electron microscopy and most commonly, serology.

3.4. MORPHOLOGY

Plant viruses come in different shapes and sizes nearly half of them are elongated (rigid rods or threads), and almost as many are spherical (isometric)

Each plant virus consist of at least of nucleic acid and a protein.

3.5. TRANSMISSION

Plant viruses enter cells only through wounds made mechanically (by human machines such as tractors, hoes, etc) or by vectors or by deposition into an ovule by an infected pollen grain.

When a virus infects a plant, it moves from one cell to another and multiplies in most, if not all of such cells.

Viruses move from cell to cell through the plasmodesmata (small channels that directly connect the cytoplasm of neighbouring plant cells to each other and therefore establishes living bridges between cells) connecting adjacent cells.

DATE: / /

The most obvious symptoms of virus-infected plants usually appear on the leaves
In almost all virus diseases of plant occuring in the field, the virus is present throughout the plant (systemic infection) and induces the formation of systemic symptoms.
It is important to understand the many viruses infect

3.7 CLASSIFICATION OF PLANT VIRUSES

Many plant viruses are named after the most conspicuous (visible) symptoms they cause on the first host in which they have been studied. Thus, a virus causing a mosaic on tobacco is tobacco mosaic virus (TMV), whereas the disease itself is called tobacco mosaic; another virus causing spotted wilt ~~virus~~ symptoms on tomato is called tomato spotted wilt virus (TSWV) and the disease is called tomato spotted wilt, and so on.

All viruses belong to the kingdom viruses. Within the kingdom, viruses are distinguished as RNA viruses and DNA viruses, depending on whether the nucleic acid of the virus is Ribonucleic acid (RNA) or Deoxyribonucleic acid (DNA).

Viruses are further sub-divided depending on whether they possess one or two strands of RNA

or DNA of either positive or negative sense, either filamentous or isometric.

3.8. CONTROL OF PLANT VIRUSES

The best way to control a virus disease is by keeping it out of an area through systems of quarantine (isolation), inspection (via plant-phyto-sanitary servicing department) and certification
Eradication (elimination) of diseased plants to eliminate inoculum (portion of infection) from the field may, in some cases, help control the disease
Plant may be protected against certain viruses by protecting them against the virus vectors
The use of virus-free seed, tubers, cutting, and so on is the single most important measure for avoiding virus diseases of many crops, especially those lacking insect vectors.

after which the virus in some of them is inactivated and the plants are completely healthy
No chemical substances (viricides) are yet available for controlling virus diseases of plants in the field, although some compounds can help reduce the problem.

3.9. EXAMPLES OF DISEASES CAUSED BY VIRUSES

I. TOBACCO MOSAIC VIRUS (TMV)

Symptoms consist of various degrees of mottling, chlorosis, curling, distortion, and dwarfing of leaves, flowers and entire plant.

On tomato leaflets may become long and pointed and, sometimes, shoestring like. Infections in young plant reduce fruit set.

II. BEAN COMMON MOSAIC AND BEAN YELLOW MOSAIC

Both Bean Common Mosaic and Bean Yellow Mosaic occur wherever beans is grown. Bean Common Mosaic affects all beans, but only beans (Phaseolus vulgaris and some other Phaseolus spp.), whereas Bean Yellow Mosaic also affect peas.

They are often found in the same field and often on the same plants. Plants infected with either virus may show mottling yellowing and malformation of leaves and pods.

Yields may be reduced by up to 80 to 100% depending on the plant stage of growth at the time of infection.

Mottling: An irregular arrangement of patches or spots of colour.

III. POTATO VIRUS Y (PVY)

Potato virus Y occurs worldwide and is of great economic importance (reduce yield by wiping out almost everything).

It affects potato, pepper, tomato and tobacco and causes severe losses on all these hosts

Symptoms vary from a mild to severe mottle on most hosts to a streak or "leafdrop streak", resulting from long necrotic lesions along the veins on the underside of leaflets of some potato varieties.

It is transmitted mainly through infected potato seed tubers.

IV. POTATO LEAFROLL (PLR)

Potato leafroll occurs worldwide. It is caused by the Potato Leafroll virus (PLRV) and affects potatoes.
It causes high yield losses and can be the most devastating virus of potato.
It causes prominent upward rolling of the leaves, and the plants are stunted and have a stiff upright growth.
The virus is transmitted through infected potato seed tubers and, in the field by more than 10 species of aphids in the persistent manner.
Its control depends on the use of potato seed tubers free of the virus.
Because the vector must feed for several hours to acquire the virus and for several more hours to infect the plant with the virus, some of control of PLRV has been achieved through early control of the aphid vectors with insecticides.

V. MAIZE STREAK VIRUS (MSV)

Maize streak is widespread and severe in the Southern half of Africa.
It appears as spots on the leaves that later develop into streaks (patches of colour).
Plants infected are stunted and their ears are poorly filled. Ears : middle part of cob
The maize streak geminivirus is transmitted by several species of leafhoppers.

VI. CASSAVA MOSAIC VIRUS (CMV)

The disease is present wherever cassava is grown. Cassava plants affected with Cassava moisaic virus produces leaves that exhibit mild to severe mosaic symptoms.
The leaf blades become narrow, entire plants remain stunted and forms small tubers that contain only the small or little amount of starch, especially in younger plants.
In older plants the symptoms are mild and more severe in young cassava.

TRANSMISSION OF CMV

CMV is transmitted through planting of infected cassava plantings or stems which issue is used as propagated materials (vector) for CMV
Vectors for Cassava mosaic virus are whiteflies

PREVENTION

Planting disease-free cutting or seeds
Isolation when there is an infection E.g transferring of soil portion from an infected plant from field to field or transfer of infected plant itself.
Spray the chemicals to avoid (reduce) the vectors.

DISEASES CAUSED BY VIROIDS

Viroids are similar to viruses but viroids are much more smaller, low molecular weight RNA and they also infect plant cells, replicate themselves and cause diseases.
RNA in virus is enclosed in proteins but RNA in viroids exist as free or naked RNA.
Diseases caused by viroids are not common, and how they replicate themselves is still not yet known.

TRANSMISSION OF VIROIDS

Viroids are spread from diseased to health plants mainly through mechanical means (through human machines like rakes, hoes, tractors by ploughing propagating, grafting, etc.)
Several viroids such as Potato spindle tuber or tomato bunch top are also transmitted through pollens and seeds.
Viroids can also be transmitted by vectors on their mouth part or feet of some insects.

EXAMPLES OF VIROIDS

Potato Spindle Tuber Viroid

The first to be recognized by man. It causes severe losses and it can attack all the varieties of potato
It spreads rapidly.

Symptoms of SPTV

Infected plant (potato plant) appear erect and dwarf
The leaves as well are small and erect and show signs of rolling and twisting.
The tubers are small, elongated with tapering ends

DATE: / /

The tubers have the eyes that are more conspicuous (visible) and numerous.

Control of Potato Spindle Tuber Viroid

Sanitation : Cleaning of tools, proper weeding
Planting disease free seeds, cutting
Ensuring that the soils are free from diseases

* How do insects develop resistance as a mechanism to pesticides ???

Insects possess perculiar (rare) genes that enables them to withstand or survive adverse (harsh) conditions even after spraying pesticides.
For instance, out of 100%, 97% dies and the remaining 3% starts replicating, producing another perculiar gene insects reducing yields by up to 80 to 100% of production.
Therefore, farmers should develop a habit of changing chemicals to keep pests at low rates on farms when pests withstands chemical spray.

4.0 NEMATODES

4.1 INTRODUCTION

Nematodes belong to the kingdom Animalia. They are wormlike in appearance but quite distinct taxonomically from the true worms.

Most of the several thousand species of nematodes live freely in fresh or salt waters or in the soil and feed on micro-organisms and microscopic plants and animals.

They are extremely small and as such can't be seen with the naked eye but they can be observed under a microscope.

The nematode body is more or less transparent and translucent (clear).

Several species of nematodes are known to feed on living plants, obtaining their food with spears or stylets and causing a variety of plant diseases worldwide.

4.2. ECOLOGICAL CLASSIFICATION OF NEMATODES

There are two major classes

I. Above ground feeder

II. Below ground feeder

4.2.1 ABOVE GROUND FEEDER

a. Feeding on flower buds, leaves and bulbs

i. Seed gall nematode ; Anguina tritici

ii. Leaf and bud nematode ; Aphelenchoides

iii. Stem and bulb nematode ; Dictylenchus

b. Feeding on tree trunk

i. Red ring nematode ; Rhadinaphelenchus cocophilus

ii. Pine wilt nematode ; Bursaphelenchus xylophilus

DATE: / /

4.2.2. BELOW GROUND FEEDER

It is again classified into three classes

I. Endoparasitic nematodes

II. Semi-endoparasitic nematodes

III. Ectoparasitic nematodes

4.2.2.1. ENDOPARASITIC NEMATODES

The entire nematode is found inside the root and the major portion of nematode body found inside the plant tissue. They are two types:

a. Migratory endoparasite

These nematodes move in cortical parenchyma of host root. While migrating they feed on cells, multiply and cause necrotic lesion. Examples, Pratylenchus spp., Radopholus spp. and Hirschmanniella spp.

b. Sedentary endoparasite

Sedentary: Remaining in one place, especially when firmly attached to some object.

The second stage larvae penetrate the rootlets and become sedentary throughout the life cycle inside the root cortex. Examples, Heterodera spp. and Meloidogyne spp.

4.2.2.2. SEMI-ENDOPARASITIC NEMATODES

The anterior part of nematode, head and neck being permanently fixed in the cortex and the posterior part extends free into the soil. Examples, Rotylenchulus reniformis and Tylenchulus semipenetrans

4.2.2.3 ECTOPARASITIC NEMATODES

These nematodes live freely in the soil and move

closely or on the root surface, feed intermittently on the epidermis and root hairs near the root tip. They are two types:

a. Migratory ectoparasites
These nematodes spend their entire life cycle free in the soil, feed externally on the host plants, deposit eggs in soil. When the roots are disturbed they detach themselves. Examples, Criconemoides spp., Paratylenchus spp., etc.

b. Sedentary ectoparasites
In this type of parasitism the attachment of nematode to the root system is permanent but for is similar to the previous one

PARASITISM AND HABITAT

i. Females and III and IV stage of larvae are sendentary endoparasites.
ii. Males and II stage larvae are migratory

MORPHOLOGICAL CHARACTERS

i. Body
Elongate larvae and male typically saccate, spheroid with a distinct neck in females.
ii. Stylet
In males, strong with rounded knob and in females, more slender than males.
iii. Oesophagous
With large median bulb followed by short isthumus.

iv. Excretory pore
Often seen with part of excretory tube in the area between posterior part of stylet knobs and opposite to the median bulb.

v. Vulvas and anus
Female typically opposite to neck and surrounded by a pattern of fine lines like human finger print (Perennial pattern)

vi. Spicule
, Very near the terminus of males Bursa is absent.
, Yellowing of leaves
, Stunted growth
, Reduced vigor (capability to grow)
, Reduced size and number of fruits
, Gall formation
, Multinucleate cell
, Giant cell (Nurse cell)
, Hypertrophy - Enlargement of cell
, Hyperplasia - Multiplication of cell

CONTROL

Two to three deep ploughing
Rotation with cereal crops
Application of nematocides. Apply carbofuron (Furdan 3G) at 7g/m
Resistant varieties of tomato (or any other crop)

SYMPTOMS CAUSED BY NEMATODES

Most of the plant parasitic nematode affect the root portion of plants except:
Anguina spp.

Aphelenchus spp.
Aphelenchoides spp.
Ditylenchus spp.
Rhadinaphelenchus cocophilus and
Bursaphelenchus xylophilus

Nematode suck the sap of the plant with the help of stylet and causes leaf discolouration, stunted grow reduced leaf size and fruits and lesions on roots, gall reduced root system and finally wilting.

Symptoms of nematode disease can be classified as

A. Symptoms produced By Above Ground Feeder Nematodes

i. Dead or devitalized buds
Nematode infection kills growing buds e.g. Aphelencho fragariae on strawberry.

ii. Crinkled stems and foliage
Eg. Wheat gall nematode, Anguina tritici ~~larva~~ Ufra disease of rice, Ditylenchus angustus.

iii Seed galls
Eg. Wheat gall nematode, Anguina tritici larva ente into the flower primordium and develops a gall.

iv. Necrosis and discolouration
, Red ring disease of coconut, Rhadinaphelenchus cocophilus
, Due to the infestation, red coloured circular area appear in the trunk of infested palm.

v. Leaf lesions
, Symptom on broad

ABOVE GROUND SYMPTOMS

i. Stunting:

- Reduced plant growth and the plant cannot be able to withstand in adverse conditions. Patches on stunted plants appears in the field. E.g. Heterodera arenae.
- Molya disease in wheat and barley. E.g. Globodera rostochiensis
- Golden nematode in potato

ii. Discolouration of foliage: Also due to nutritional deficiency. eg:
Root lesion nematode, Pratylenchus coffeae,
White tip nematode, Aphelenchoids basseyl,
Citrus nematode, Tylenchulus semipenetrans.

iii. Wilting:
Eg. Root-knot nematode. Meloidogyne spp.

iv Decline and die back:
Die back (dying backwards, from leaf tip to the rest of the leaf blade). Eg. in banana decline and die back are caused by Radopholus

BELOW GROUND SYMPTOMS

i. Root Galling:
E.g Meloidogyne spp., Xiphinema diversicaudatum

ii Reduced Root System:
Due to nematode feeding the root tip growth is arrested and the root produce branches. This may be of various kinds such as coarse root, stubby root and curly root
a. Stubby roots; Stubby branches or rootlets arranged in cluster. Eg. Trichodorus christiei on corn.
b. Coarse roots; Lateral roots stopped growth with no branches eg. Belonolaimus longicaudatus on corn
c. Curly roots; The nematode retard the elongation of roots and cause curling of roots known as 'Fish hook' symptom. Eg. Injury caused by Xiphinema spp.

iii. Root lesions:
Necrotic lesions e.g Pratylenchus spp. (soybean), Radopholus similis (citrus and banana), Helicotylenchus multicinctus (banana).

iv. Rotting:
Nematode + micro-organisms Eg Bitylenchus destructor
Potato rot

v. Excessive root branching:
Eg. Meloidogyne hapla in tomato

* Nematode infections of plants result in the appearance of symptoms on roots as well as or above ground parts of plants.
* Root systems may appear as root lesions, root knots or root galls.
* Excessive root branching, injured root tips, and when nematode infections are accompanied by plant pathogenic or saprophytic bacteria and fungi, as root rots.
* Some species of nematodes attacks upper part of the plant and cause twisting or distortion of leaves and stems,

DATE: / /

and abnormal development of the floral parts
* Certain nematodes attack cereals or grasses and form galls full of nematodes in place of seeds.

HOW NEMATODES AFFECT PLANTS
* Most of the damage by nematodes seem to be caused by a secretion of saliva injected into the plants while the nematodes are feeding.
* Some nematode species are rapid feeders. They puncture a cell wall, inject saliva into the cell, withdraw part of the cell contents, and move on within a few seconds.
* Others feed much more slowly and may remain at the same puncture for several hours or days. These as well as the females of species that become established in or on roots permanently, injected saliva intermittently as long as they are feeding.
* The feeding process causes the affected plant cells to react, resulting in dead or devitalized root tips and buds, lesion formation and tissue breakdown, swellings and galls of various kinds, and crinkled and distorted stems and foliage.

METHODS OF DETECTING PLANT DISEASES SCIENTIFIC

* There are direct and indirect methods of detecting pl diseases
* Direct detection includes molecular and ~~cellular~~ serol serological methods that could be used to analyse large numbers of samples
* Disease causing pathogens such as viruses, bacter and fungi are directly detected ~~to idea~~ and are ~~directly~~ accurately identified and a disease is det
* Selological is Pairing of a pathogen or a piece of a speciment with anti

* Indirect way of disease identification is through using various parameters such as morphological features, symptoms based on a type of disease and so on.

DIRECT DETECTION METHODS

I. Polymerase Chain Reaction (PCR)

This is the common molecular biology technique that is used to make multiple copies of the specific DNA. A region or a portion of the infected plant is isolated and taken to the lab. The DNA in this portion will then be copied and multiplied and the results are ~~matche~~ matched to the type of disease that are there.

Polymeraze chain reaction (PCR) is a common molecular biology technique that enables researcher to make multiple copies of the specific region of DNA. PCR is efficient, rapid and can amplify DNA or RNA sequences from various sources. Once the DNA has been sufficiently amplified, the resulting product

be sequenced, analyzed by gel electrophoresis, or cloned.

PCR was initially used for highly specific detection of diseases caused by bacteria and viruses but now it has widely been used for the detection of plant pathogens as well. i.e fungi and nematodes too.

Guide on how PCR is performed for plant disease detection:

1. Sample Collection: Collect plant tissue (leaves, stems, roots) showing disease symptoms.
   *Ensure proper storage (e.g. freezing at -20°C or using RNA/DNA stabilization buffers).
2. Nucleic Acid Extraction:
   a. DNA Extraction (for bacteria, fungi, DNA viruses)
      *Grind tissue in liquid nitrogen or extraction buffer
      * Use commercial kits (e.g., CTAB method, silica-column-based kits) to isolate DNA.
      * Quantify DNA using a spectrophotometer (NanoDrop)
   b. RNA Extraction (for RNA viruses)
      * Use TRIzol or RNA-specific kits to prevent degradation.
      * Treat with DNase to remove contaminating DNA.
      * Convert RNA to cDNA using Reverse Transcriptase (RT-PCR).
3. PCR Setup
   a. Conventional PCR

* Primer design : Use pathogen-specific primers (e.g established literature or database)
* Reaction mix :
  • Temperate DNA, Forward and Reverse Primers
  • Taq DNA polymerase, PCR Buffer (with $MgCl_2$
  • Nuclease-free water
* Thermal Cycling Conditions : Temperature, time and cycles of the reactions involved.

b. Quantitative PCR (qPCR) / Real-Time PCR
* Uses fluorescent dyes (SYBR Green) or Probes (TaqMan) for quantification
* Detects pathogen load in real-time, more sensitive than conventional PCR.

c. Multiplex PCR
* Detects multiple pathogens in one reaction using different primer sets.

4. PCR Product Analysis :
a. Gel Electrophoresis :
* Run PCR product on 1.5-2% agarose gel with a DNA ladder.
* Stain with ethidium bromide (EtBr) or safer alternatives (SYBR safe).
* Visualize under UV light - a band of expected size confirms pathogen presence.

b. Sequencing (for confirmation)
* Purify PCR product and send for sequencing
* Compare with databases (NCBI BLAST) for pathogen identification

5. Interpretation of Results
a. Positive Control : Known pathogen DNA should amplify
b. Negative Control : No template (water) should show no band
c. Sample : If band matches expected size, pathogen is present.

DATE: / /

ADVANTAGES OF PCR

* High sensitivity (detects low pathogen levels)
* Specificity (distinguishes closely related pathogens)
* Fast results (within hours)
* Works on latent infections (before symptoms appear)
* It's a common technology, portable (specimen can come from various sources) and easy to operate.
* It is accurate and more efficient (reliable, rapid and sensitive)
* It is versatile (works for viruses, fungi, bacteria and nematodes)

DISADVANTAGES OF PCR

* Requires DNA/RNA extraction (more complex)
* Risk of contamination (False positives if lab protocols aren't strict.)
* Expensive equipment (Thermal cyclers, gel electrophoresis setups)
* Does not confirm viability (Detects dead or inactive pathogens

LIMITATIONS OF PCR

* Its effectiveness is subjected to DNA extraction, inhibitors, polymerase activity and concentration of PCR buffer.

II. Fluorescence In-Situ Hybridization (FISH)

This another type of molecular detection technique which is applied for bacterial detection in combination with microscopy and hybridization of DNA probes and target gene from plant samples.

Due to the presence of pathogen-specific ribosomal RNA (rRNA) sequences in plants, recognizing this specific information by FISH can help detect the pathogen infections in plants. In addition to bacterial pathogens, FISH could also be used to detect fungi and viruses and other endosymbiotic bacteria that infect the plant.

Applications in Plant Pathology

* Detects bacteria (e.g. Candidatus Liberibacter in citrus greening)
* Identifies fungal infections (e.g., Fusarium spp.)
* Visualizes viral RNA in infected cells

Steps in FISH for Plant Disease Detection

1. Sample Preparation
   * Fix plant tissue in formaldehyde or ethanol to preserve nucleic acids
   * Embed in paraffin or cryosections for thin slicing
   * Permeabilize cells with enzymes (e.g. cellulase, pectinase) to allow probe entry
2. Probe Design
   * Probes (~20-30 nucleotides) are designed to bind pathogen-specific bacteria or RNA/DNA sequences
   * Probes are labeled with fluorophores (e.g FITC)
3. Hybridization
   * Apply probe to tissue sections.

* Incubate at 37 - 55°C (depending on probe specificity)
* Wash off unbound probes to reduce background noise.

4. Detection and Microscopy
* View under a fluorescence microscope or confocal laser microscope.
* Pathogen signals appear as bright spots (e.g bacterial cells glowing red / green)

ADVANTAGES OF FISH
* It is very high sensitivity.
* Visual localization of pathogens inside plant tissues.
* High specificity (species - level detection)
* Works for unculturable microbes (eg phytoplasma
* No need for DNA extraction - Probes bind directly in tissues
* Multiplexing possible - Different probes with different colours.

DISADVANTAGES OF FISH
* Expensive microscopy; requires fluorescence / confocal microscopes
* Complex protocol; fixation, permeabilization and hybridization steps
* Low throughput; time - consuming sample Preparation (labor - intensive sample preparation)

III. Enzyme-Linked Immunosorbent Assay (ELISA)

The Enzyme-linked immunosorbent assay (ELISA) another molecular method for identification of disease based on antibodies and colour change in the assay (attempt).

In this method, the target epitopes (the site on the surface of an antigen molecule to which an antibody attaches itself) from the viruses, bacteria and fungi are made to specifically bind with antibodies conjugated (coupled) to an enzyme.

The detection can be visualized based on colour changes resulting from the interaction between the substrate (from the lab) and the immobilized enzyme (extracted from the plant).

Types of ELISA used in Plant Pathology

Direct ELISA - primary antibody is enzyme-linked.

Indirect ELISA - secondary antibody is enzyme-linked (more sensitive).

DAS-ELISA (Double Antibody Sandwich ELISA) - common for plant viruses.

Steps in ELISA for Plant Disease Detection

1. Sample Preparation
   * Grind plant tissues in extraction buffer
   * Centrifuge to remove debris
2. Coating the Plate
   * Coat wells with capture antibody (specific to the pathogen)
   * Incubate at 4°C overnight or 37°C for 1-2 hours
   * Wash with PBS-Tween to remove unbound antibodies.

3. Adding the sample
* Load plant extract into wells
* Incubate (1-2 hours) to allow antigen-antibody binding.
* Wash to remove unbound material

4. Detection
* Add enzyme-linked secondary antibody
* Incubate and wash
* Add substrate
* Colour develops if pathogen is present.

5. Measurement
* Read absorbance using a microplate reader
* Higher absorbance = more pathogen present

Advantages of ELISA
* Low cost, visual color change can be used for detection
* Cost effective compared to PCR for large screening
* No need for DNA extraction
* High-throughput (many samples at once

Disadvantages of ELISA
* It has low sensitivity for bacteria
* Requires specific antibodies (not always available for new pathogens.
* Cross-reactivity - False positives if antibodies bind non-target proteins
* No genetic info - cannot identify pathogen strains (proteins) like PCR.

N.B on Pros: Simple & rapid (Results in hours) Detects proteins (useful for pathogens where DNA is hard to extract

PCR vs. ELISA vs. FISH

Feature	PCR	ELISA	FISH
Target	DNA / RNA	Proteins (antigens)	DNA / RNA tissues
Sensitivity	Very High	Moderate	Moderate
Speed	Fast (2-4 hrs)	Fast (2-3 hrs)	Slow (sa... Prep-heavy
Throughput	Medium (batch testing)	High (96-well plates)	Low (1-2 Samples/sl...
Cost	High (equipment or reagents)	Low-medium	High (microscop... or probes)
Best use	Early detection, strain ID	Large-scale screening	Pathogen localizatio...

INDIRECT DETECTION METHODS

These are methods based on the features being expressed by the plants such as

* leaf size,
* discolouration
* Molting or stunted growth
* wilting

using optical sensors to detect ~~to~~ biotic and abiotic stress in plant etc

REMOTE SENSING (RS)

Remote sensing is estimating an object / phenomenon without being in physical contact with it.

Remote sensing is a science / art that permits us to obtain information about an object / a phenomenon through analysis of data obtained through sensory devices with being in physical contact with that object.

OBJECTIVES OF REMOTE SENSING IN PLANT PATHOLOGY

1. Assessment of disease over a vast area.
2. To know the relationship of diseases and environment.
3. To know the origin and development of epidemics.
4. Quantitative assessment of the disease.

REMOTE SENSING TECHNIQUES OF IMPORTANCE TO PLANT PATHOLOGY

1. AERIAL PHOTOGRAPHY

This can detect objects on land over a large area. Colwell (1956) first used remote sensing technique for monitoring stem rust of wheat. He showed that panchromatic colour and especially infrared aerial photography could be used to detect rusts and viral diseases of small grains and certain diseases of citrus. Later, infrared photography was used in England for late blight of potato.

NB: The key to distinguish diseased and healthy parts of a crop is to use appropriate film or

filter combinations. The main film types used are panchromatic, infrared, normal colour and colour infrared. The infrared films are preferred because of their superior sensitivity to visible light and near infrared wavelengths of radiation (700-900m The colour infraied or Ektachrome Aero Infrared (Camouflage Detection Film) is superior as it can show the difference between diseased and healthy patches of plants in colour. The healthy foliage is highly reflective to the infrared wavelengths and appears red on this film whereas blighted or diseased foliage has low infrared reflectance and does not appear ~~in~~ red in the photograph.

Main film types

Panchromatic
Infrared (Superior sensitivity to visible light)
Normal colour
Colour infrared or Ektachrome Aero Infrared

2. SATELLITE IMAGING

Weather satellites

Often cyclones create heavy clouds with rains and anti-cyclone creates a cloudless sky. All these can be effectively monitored by weather satellites. Sequential pictures show the movement of these systems before they arrive in an area. Therefore monitoring epidemic favouring systems using a satell
• the disease occurrence on the field can be monitored
Example: The spread and deposition of stem rust pathogen of wheat is influenced by definite synop weather conditions called Indian stem rust rules

DATE: / /

Earth Resources Technology Satellite (LANDSAT, 1972, USA)

LANDSAT covers the entire globe every 18 days scanning the same area at a fixed time. The scanned data is compared for any major differences happened within 18 days.

Nagarajan utilized LANDSAT infrared spectral bands 6 (0·7 - 0·8 Mm) and 7 (0·8 - 1·1 Mm) to differentiate healthy wheat crop of India and severe yellow rust affected crop of Pakistan.

Examples: Coconut root rot and wilt,
Black stem rust of wheat
Citrus canker

ADVANTAGES OF REMOTE SENSING

Reveals pattern of disease incidence, intensity and development over large area.
Data generated by remote sensing is amendable to multidisciplinary approach.
Gives synoptic view of large areas.
Data generated is on a permanent scale and is unbiased.
Data acquisition is fast compared to traditional methods and data analysis is effectively utilized.
Satellite data (ERTS) obtains information of an area periodically so that the information can be updated.
It frequently poses questions for ground investigators which cannot be generated by ground parties
RS facilitates disease forecasting (predicting the outbreak of a plant disease before it comes)

Non-destructive: Gathers data without disturbing plants or requiring sample collection.
It informs targeted spraying, irrigation adjustments and resource allocation.
Early detection of diseases: Detects physiological stress in plants (e.g chlorosis, wilting) before visible symptoms appear, allowing for early intervention.
Time and labour efficient: Reduces the need for field labour and repeated manual inspections, especially in large farms.

DISADVANTAGES OF REMOTE SENSING

High initial cost: Equipment like drones, hyperspectral sensors, and satellite imagery can be expensive.
Need for Technical expertise: Requires trained personnel to operate equipment, process data and interpret results.
Environmental Interference: Cloud cover, shadows and atmospheric conditions can reduce image clarity and accuracy.
Limitations of Resolution: Some sensors may not detect diseases at early stages or on individual plants especially in dense canopies.
Data overload: Large volumes of data require substantial storage and advanced analysis tools.
Limited Disease Specificity: Remote sensing often detects stress, not specific pathogens - further ground truthing is needed to confirm disease cause.
Access and infrastructure issues: May be less practical in remote or low-resource areas with poor internet and electricity access.

INTEGRATED PEST AND DISEASE MANAGEMENT (IPDM)

INTRODUCTION

IPDM involve management systems which utilize compatible combinations of all the available techniques to keep the pathogen population below the economic threshold level (ETL) which would not result in economically unacceptable damage to the crop.

IPDM is based on six principles of plant disease management and integrates multi-disciplinary approaches for the management of plant diseases.

MAIN COMPONENTS OF IPDM

1. CULTURAL PRACTICES

These involve modifying farming practices to make the environment less favourable for pests and diseases.

Examples: Crop rotation, intercropping, timely planting, proper spacing, sanitation (proper weeding, cleaning of tools), and use of resistant varieties.

2. REGULATORY MEASURES (QUARANTINE)

Legal and policy-based actions to prevent the introduction and spread of pests and diseases

Examples: Inspection of imported plant materials, destruction of infested materials, restriction on movement of infected plants.

3. CHEMICAL METHODS

Use of pesticides (fungicides, insecticides, herbicides) to control pests and pathogens

should be used judiciously to avoid resistance, environmental harm and health risks.

4. BIOLOGICAL METHODS

Use of natural enemies like predators, parasitoids, or microbial agents (e.g., Bacillus thuringiensis) to manage pests and diseases.
It is environmentally friendly and promotes ecological balance.

5. PHYSICAL METHODS

Use of mechanical or physical tools to manage pests and diseases.
Examples: Hand-picking, traps, barriers, heat treatment, solarization (overexposure to sunlight), and proper drying/storage techniques.

6. GENETIC ENGINEERING

Involves modifying the genetic makeup of crops to make them resistant to specific pests or diseases.
Example: Bt cotton (resistant to bollworms), or virus-resistant genetically modified papaya.

MAIN STRATEGIES OF IPDM

1. Need based application of pesticides.
2. Encouragement and enhancement of biocontrol agents.